Operational guide to network_monitor2

This guide provides the minimum procedure for selecting a network path and running a test between NUC2 and NUC1. See README.md for installation, database configuration, and advanced options.

To run one test with custom duration, protocol, bandwidth, packet size, flow count, or direction, see [docs/MANUAL_TEST_GUIDE_EN.md](../MANUAL_TEST_GUIDE_EN.md).

1. Before you begin

> Safety: never modify or interrupt the management connection on VLAN27.

- NUC2 management: 10.10.27.11, interface enp114s0.
- NUC1 management: 10.10.27.10, interface enp114s0.
- Default gateway: 10.10.27.1 on enp114s0.
- Connect to NUC1 through ssh 10.10.27.10.

Do not change VLAN27, the management addresses, or the default route.

Check the tool on both NUCs:

```
command -v network_monitor2
network_monitor2 --help
command -v iperf3
```

2. Select the path

- CEIT: NUC2 10.10.42.11 → NUC1 10.10.22.10.
- INDRA: NUC2 10.10.42.11 → NUC1 10.10.22.10.
- Bypass TN0: NUC2 10.10.28.11 → NUC1 10.10.28.10.
- Bypass TN1: NUC2 10.10.29.11 → NUC1 10.10.29.10.

CEIT and INDRA use the same endpoint addresses but different routes and gateways. The bypass paths are directly connected.

CEIT

From NUC2, select CEIT on both NUCs:

```
sudo /home/tuser/scripts/test_ceit
ssh -t 10.10.27.10 'sudo /home/tuser/scripts/test_ceit'
```

```
# NUC2: must show via 10.10.42.41
ip route get 10.10.22.10 from 10.10.42.11

# NUC1: must show via 10.10.22.40
ssh 10.10.27.10 \
'ip route get 10.10.42.11 from 10.10.22.10'
```

INDRA

From NUC2, select INDRA on both NUCs:

```
sudo /home/tuser/scripts/test_indra
ssh -t 10.10.27.10 'sudo /home/tuser/scripts/test_indra'
```

```
# NUC2: must show via 10.10.42.31
ip route get 10.10.22.10 from 10.10.42.11

# NUC1: must show via 10.10.22.30
ssh 10.10.27.10 \
'ip route get 10.10.42.11 from 10.10.22.10'
```

Bypass TN0 or TN1

There is no test_bypass script. Before using a bypass path, remove only the CEIT/INDRA test routes. These commands must not change VLAN27 or the default route:

```
# NUC2
sudo ip route del 10.10.22.0/24 via 10.10.42.41 2>/dev/null || true
sudo ip route del 10.10.22.0/24 via 10.10.42.31 2>/dev/null || true

# NUC1, through management
ssh -t 10.10.27.10 \
'sudo ip route del 10.10.42.0/24 via 10.10.22.40 2>/dev/null || true; \
sudo ip route del 10.10.42.0/24 via 10.10.22.30 2>/dev/null || true'
```

Check the selected path:

```
# TN0
ip route get 10.10.28.10 from 10.10.28.11

# TN1
ip route get 10.10.29.10 from 10.10.29.11
```

The result must show a directly connected route with no gateway. To return to CEIT or INDRA, rerun the corresponding script on both NUCs.

3. Check reachability

From NUC2, run the ping for the selected path:

```
# CEIT or INDRA
ping -I 10.10.42.11 -c 5 10.10.22.10

# Bypass TN0
ping -I 10.10.28.11 -c 5 10.10.28.10

# Bypass TN1
ping -I 10.10.29.11 -c 5 10.10.29.10
```

If ping reports 100% packet loss, do not run the throughput test. Check routes, addresses, VLANs, and ARP first.

4. Run a TCP test

Two terminals are required. Start the server on NUC1 first, then the client on NUC2. This example uses TN0.

Terminal 1 — server on NUC1

```
ssh 10.10.27.10
network_monitor2 --server \
--bind-ip 10.10.28.10 \
--port 5050
```

Leave the server running.

Terminal 2 — client on NUC2

```
network_monitor2 10.10.28.10 \
--source-ip 10.10.28.11 \
--port 5050 \
--duration 30 \
--tcp
```

Monitoring run completed means that throughput, ping, and interruption monitoring have finished. The client stores results in NUC2's local database; the server stores its own iperf3 samples locally.

When finished, stop the server with Ctrl+C.

5. Adapt the command to the path

Only the addresses change:

- CEIT or INDRA: server 10.10.22.10, client/source 10.10.42.11.
- TN0: server 10.10.28.10, client/source 10.10.28.11.
- TN1: server 10.10.29.10, client/source 10.10.29.11.

Always use --bind-ip on the server and --source-ip on the client. --interface is optional and acts as an additional hint or metadata.

6. UDP test

Leave the same server running, use --udp, and specify the offered traffic rate:

```
network_monitor2 10.10.28.10 \
--source-ip 10.10.28.11 \
--port 5050 \
--duration 30 \
--udp \
--bandwidth 400M
```

For UDP, interpret received bitrate, jitter, and packet loss together. High bitrate with high loss does not indicate a stable link.

7. Test from NUC1 to NUC2

Reverse the server and client. TN0 example:

On NUC2:

```
network_monitor2 --server --bind-ip 10.10.28.11 --port 5050
```

On NUC1:

```
network_monitor2 10.10.28.11 \
--source-ip 10.10.28.10 \
--port 5050 --duration 30 --tcp
```

8. View the results

Terminal

For TCP, check throughput and retransmissions (Retr). For UDP, check received bitrate, jitter, and packet loss.

Excel

Each NUC exports its own local database:

```
network_monitor2 --export-excel results.xlsx
```

To restrict the time interval:

```
network_monitor2 --export-excel results.xlsx \
--start "2026-09-03 09:00:00" \
--end "2026-09-03 12:00:00"
```

Grafana

Open http://<GRAFANA_IP>:3000 and select the local or bidirectional dashboard. Check the time range if no data appears.

9. Automated KPI campaign

From NUC2, preview it first:

```
cd /home/comsa/network_monitor_v2
./run_kpi_tests.sh --dry-run
```

Then start the campaign:

```
./run_kpi_tests.sh
```

The current campaign tests CEIT, bypass TN0, and bypass TN1, writes reports and logs under results/, and restores the initial route state after success, failure, or Ctrl+C. INDRA is not yet included in the automated campaign.

This campaign is separate from tests/scripts/run_nuc2_base_measurements.sh, which runs the Cedex TN0/TN1 matrix and exports its results under export/.

10. Common problems

Connection refused

Check that the server is running and that the server and client use the same port.

No route to host or no ping response

```
ip route get <DESTINATION> from <SOURCE>
ip neigh
ping -I <SOURCE> -c 5 <DESTINATION>
```

A FAILED neighbor normally indicates an ARP, VLAN, or Layer 2 problem.

Port already in use

Select another port, such as 5051, and use it on both the server and client.

No data in Grafana

Check the time range, /opt/network_monitor2/setup.conf, and whether the test displayed Monitoring run completed.

Quick reference

```
# Server
network_monitor2 --server --bind-ip <LOCAL_IP> --port 5050

# TCP client
network_monitor2 <SERVER_IP> --source-ip <CLIENT_IP> \
--port 5050 --duration 30 --tcp

# UDP client
network_monitor2 <SERVER_IP> --source-ip <CLIENT_IP> \
--port 5050 --duration 30 --udp --bandwidth 100M

# Local export
network_monitor2 --export-excel results.xlsx
```